

32.6 Bayesian thread III: mixed models as hierarchical models

The Bayesian thread began in [Section 7.6](#) with a conjugate prior on $(\boldsymbol{\beta}, \sigma^2)$ and continued in [Section 12.6](#) with credible sets and predictive distributions. Mixed models are where the two traditions meet most closely, because a random effect *is* a prior: the statement $\mathbf{u} \sim N_q(\mathbf{0}, \mathbf{G})$ is formally indistinguishable from a normal prior on q unknown constants. What separates the methods of [Section 32.4](#) from a fully Bayesian analysis is only that they estimate $\mathbf{G}$ and then treat it as known, which is empirical Bayes (Proposition 27.5.4), while a Bayesian puts a prior on $\mathbf{G}$ too and integrates.

32.6.1 The hierarchical reading

Write the model in three levels:

$$\mathbf{Y} \mid \boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta} \sim N_n(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}, \mathbf{R}), \quad \mathbf{u} \mid \boldsymbol{\theta} \sim N_q(\mathbf{0}, \mathbf{G}), \quad (\boldsymbol{\beta}, \boldsymbol{\theta}) \sim \pi. \quad (32.6.1)$$

Level one is a linear model with known coefficients; level two is the prior on $\mathbf{u}$ that the experiment's sampling of levels supplies; level three is the part that is genuinely a matter of choice.

Proposition 32.6.1 — BLUP as a posterior mean, REML as a marginal posterior

Consider [Eq. 32.6.1](#) with $\mathbf{G}$ and $\mathbf{R}$ positive definite and $\text{rank}(\mathbf{X}) = p$.

- (a) (*BLUP is a posterior mean.*) Fix $\boldsymbol{\theta}$ and take the improper flat prior $\pi(\boldsymbol{\beta}) \propto 1$. Then the joint posterior of $(\boldsymbol{\beta}, \mathbf{u})$ is

$$\begin{pmatrix} \boldsymbol{\beta} \\ \mathbf{u} \end{pmatrix} \mid \mathbf{y} \sim N_{p+q} \left(\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{pmatrix}, \mathbf{C}^{-1} \right),$$

with $\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}$ and $\mathbf{C}$ as in Theorem 32.3.2. In particular $E(\mathbf{u} \mid \mathbf{y}) = \hat{\mathbf{u}}$ is the BLUP, $E(\boldsymbol{\beta} \mid \mathbf{y}) = \hat{\boldsymbol{\beta}}$ is the generalized least squares estimate, and $\text{Cov}(\mathbf{u} \mid \mathbf{y})$ is the prediction error covariance of [Eq. 32.3.2](#).

- (b) (*REML is a marginal posterior.*) With the flat prior on $\boldsymbol{\beta}$ and any prior $\pi(\boldsymbol{\theta})$, the marginal posterior of the variance components is

$$\pi(\boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \exp\{\ell_R(\boldsymbol{\theta})\}, \quad (32.6.2)$$

with ℓ_R the restricted log-likelihood of Definition 32.4.1. The REML estimate is therefore the mode of the marginal posterior under a flat prior on $\boldsymbol{\theta}$.

- (c) (*What the plug-in leaves out.*) Under (a) with $\boldsymbol{\theta}$ replaced by an estimate, the reported posterior variance is $\text{Cov}(\mathbf{u} \mid \mathbf{y}, \hat{\boldsymbol{\theta}})$, whereas the correct marginal posterior variance is $E\{\text{Cov}(\mathbf{u} \mid \mathbf{y}, \boldsymbol{\theta}) \mid \mathbf{y}\} + \text{Cov}\{E(\mathbf{u} \mid \mathbf{y}, \boldsymbol{\theta}) \mid \mathbf{y}\}$. The plug-in omits the second term, so it is too small.

Proof.

- (a) With $\pi(\boldsymbol{\beta}) \propto 1$ and $\boldsymbol{\theta}$ fixed, the joint density of $(\mathbf{y}, \mathbf{u})$ times the prior is proportional to $\exp\{-\frac{1}{2}Q(\boldsymbol{\beta}, \mathbf{u})\}$ with Q the penalized criterion Eq. 32.3.6, because $-2 \log f(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{u}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})^\top \mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})$ and $-2 \log f(\mathbf{u}) = \mathbf{u}^\top \mathbf{G}^{-1}\mathbf{u}$, up to terms free of $(\boldsymbol{\beta}, \mathbf{u})$. By Theorem 32.3.2(b), Q is a quadratic in $\mathbf{w} = (\boldsymbol{\beta}^\top, \mathbf{u}^\top)^\top$ with positive definite matrix $\mathbf{C}$, minimized at $\hat{\mathbf{w}} = (\hat{\boldsymbol{\beta}}^\top, \hat{\mathbf{u}}^\top)^\top$, so $Q(\mathbf{w}) = (\mathbf{w} - \hat{\mathbf{w}})^\top \mathbf{C}(\mathbf{w} - \hat{\mathbf{w}}) + Q(\hat{\mathbf{w}})$. The posterior is therefore $N_{p+q}(\hat{\mathbf{w}}, \mathbf{C}^{-1})$, and Theorem 32.3.2(c) identifies the blocks.
- (b) The joint posterior is $\pi(\boldsymbol{\beta}, \boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \exp\{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})\}$ with ℓ the marginal log-likelihood Eq. 32.4.1, the random effects having already been integrated out in Eq. 32.2.2. Completing the square as in Lemma 32.4.1(b),

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{P}\mathbf{y} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}),$$

so integrating over $\boldsymbol{\beta} \in \mathbb{R}^p$ contributes the factor $(2\pi)^{p/2} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1/2}$ and

$$\begin{aligned} \int e^{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})} d\boldsymbol{\beta} &= \exp\left[-\frac{1}{2}\{(n-p) \log 2\pi + \log \det \mathbf{V} \right. \\ &\quad \left. + \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) + \mathbf{y}^\top \mathbf{P}\mathbf{y}\}\right]. \end{aligned}$$

By Eq. 32.4.5 the exponent is $\ell_R(\boldsymbol{\theta})$ up to the additive constant $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X})$, which does not involve $\boldsymbol{\theta}$. That is [Eq. 32.6.2](#).

- (c) This is the decomposition of a variance into the mean of the conditional variance plus the variance of the conditional mean (Proposition 2.5.4), applied to the posterior distribution with $\boldsymbol{\theta}$ as the conditioning variable. The second term is nonnegative definite. ■

Part (b) is Harville's (1974) observation, and it settles a question that the frequentist derivation leaves slightly mysterious. REML looked like a technical device for recovering lost degrees of freedom; it is in fact the exact marginal likelihood of the variance components once the fixed effects have been integrated out with a flat prior. The " $n - p$ " is not a correction but a consequence.

Idea. Three readings of the same computation. Frequentist: $\hat{\mathbf{u}}$ minimizes prediction mean squared error among unbiased linear predictors. Penalized: it minimizes a sum of squares with a ridge penalty on $\mathbf{u}$. Bayesian: it is the posterior mean of $\mathbf{u}$ under a normal prior with covariance $\mathbf{G}$. Henderson's equations compute all three at once, and the three readings differ only in what they say about $\mathbf{G}$: a covariance to be estimated, a penalty to be tuned, or a prior to be chosen.

32.6.2 Priors for variance components

The conditional distributions in [Eq. 32.6.1](#) make the conjugate choices obvious. Given $\mathbf{u}$ and $\boldsymbol{\beta}$, the sum of squares $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}\|^2$ is an ordinary normal likelihood for σ^2 , so an inverse gamma prior is conjugate; given $\mathbf{u}$, the sum $\|\mathbf{u}\|^2$ plays the same role for σ_u^2 . This is what makes the Gibbs sampler below so short. What is not obvious is which inverse gamma.

Warning. The prior $\sigma_a^2 \sim \text{IG}(\epsilon, \epsilon)$ with ϵ small is often described as non-informative. It is not, and the limit does not exist. The marginal (restricted) likelihood $\exp\{\ell_R\}$ is continuous and *strictly positive* at $\sigma_a^2 = 0$, because $\sigma_a^2 = 0$ is a perfectly good model. Any prior density whose integral diverges at the origin, such as $\pi(\sigma_a^2) \propto 1/\sigma_a^2$ — equivalently a flat prior on $\log \sigma_a$ — therefore gives an improper posterior: $\int \pi(\sigma_a^2) e^{\ell_R} d\sigma_a^2 = \infty$. For small ϵ the $\text{IG}(\epsilon, \epsilon)$ posterior is proper but arbitrarily close to improper, and with few groups the answer depends on ϵ . Gelman (2006) made this point and recommended instead a prior that is flat or half- t on the *standard deviation* σ_a , which has finite mass near zero.

A convenient weakly informative default is the half-Cauchy on σ_a with scale A , chosen large relative to the plausible size of a group effect. It keeps the Gibbs sampler conjugate through the scale-mixture representation

$$\sigma_a^2 \mid \xi \sim \text{IG}(\tfrac{1}{2}, 1/\xi), \quad \xi \sim \text{IG}(\tfrac{1}{2}, 1/A^2),$$

whose marginal for σ_a is half-Cauchy(0, A). The auxiliary variable ξ is sampled alongside everything else.

32.6.3 A Gibbs sampler in a dozen lines

For the one-way model with a flat prior on μ , the four full conditionals are all standard. Given $\mu, \sigma^2, \sigma_a^2$, the effects are independent with

$$a_k \mid \text{rest} \sim N\left(\frac{(m/\sigma^2)(\bar{y}_k - \mu)}{m/\sigma^2 + 1/\sigma_a^2}, \frac{1}{m/\sigma^2 + 1/\sigma_a^2}\right),$$

which is Eq. 32.3.3 with the current parameter values; given the effects, $\mu \sim N(\overline{y - \mathbf{Za}}, \sigma^2/n)$; and the two variances have inverse gamma conditionals with shapes $n/2$ and $g/2$ plus their prior shapes. Nothing needs tuning.

```
import numpy as np

g, m = 12, 4
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25          # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

Y = y.reshape(g, m)                          # the REML fit of Section 32.4
ms_between = m * np.sum((Y.mean(axis=1) - Y.mean()) ** 2) / (g - 1)
ms_within = np.sum((Y - Y.mean(axis=1)[:, None]) ** 2) / (n - g)
reml_s2, reml_s2a = ms_within, (ms_between - ms_within) / m

def gibbs(y, g, m, draws=40000, burn=2000, prior="inverse-gamma", A=1.0, seed=0):
    """Posterior draws of (mu, sigma^2, sigma_a^2, a) with a flat prior on mu."""
    rng = np.random.default_rng(seed)
    Y = y.reshape(g, m)
    n = g * m
    mu_, s2, s2a, xi = Y.mean(), Y.var(), 0.1, 1.0
    out = np.empty((draws, 3 + g))
    for t in range(draws + burn):
```

```

prec = m / s2 + 1 / s2a                                # a_k / rest
a = rng.normal((m / s2) * (Y.mean(axis=1) - mu_) / prec, np.sqrt(1 / prec))
mu_ = rng.normal((y - np.repeat(a, m)).mean(), np.sqrt(s2 / n)) # mu /
rest
resid = y - mu_ - np.repeat(a, m)
s2 = 1 / rng.gamma(0.001 + n / 2, 1 / (0.001 + resid @ resid / 2))
if prior == "inverse-gamma":                            # sigma_a^2 / rest
    s2a = 1 / rng.gamma(0.001 + g / 2, 1 / (0.001 + a @ a / 2))
else:                                                    # half-Cauchy(0, A)
    s2a = 1 / rng.gamma((g + 1) / 2, 1 / (1 / xi + a @ a / 2))
    xi = 1 / rng.gamma(1.0, 1 / (1 / A**2 + 1 / s2a))
if t >= burn:
    out[t - burn] = np.concatenate([[mu_, s2, s2a], a])
return out

```

[Run this code online](#)

```

# a shorter run than the 40 000 draws quoted in the text, so the cell finishes
quickly
short = gibbs(y, g, m, draws=4000, burn=500, prior="half-Cauchy", seed=12)
print(f"posterior median of sigma_a^2 {np.median(short[:, 2]):.4f}")
print(f"posterior mean of mu {short[:, 0].mean():.4f}")
print(f"REML estimate of sigma_a^2 {reml_s2a:.4f}")

```

[Run this code online](#)

Example. (The proficiency study, four ways)

Run the sampler for 40 000 draws on the data of Example (A laboratory proficiency study). Under the $\text{IG}(0.001, 0.001)$ prior the posterior median of σ_a^2 is 0.0356; under the half-Cauchy prior with $A = 1$ it is 0.0438. The REML estimate 0.0386 sits between them. The two priors therefore disagree by about a fifth of the estimate, with twelve laboratories — which is already a reason to report the prior.

The posterior means of the twelve effects track the BLUPs closely: their correlation exceeds 0.99 and the largest in absolute value is 0.294 against the BLUP's 0.321, slightly more shrunk — not because the posterior sits below the REML estimate, since its mean 0.0431 is above it, but because the shrinkage factor $B = m\sigma_a^2 / (\sigma^2 + m\sigma_a^2)$ is concave in σ_a^2 : averaging B over a posterior with mass near zero weights the data less than $\hat{\sigma}_a^2$ does. Their posterior standard deviations average 0.1218, against the plug-in prediction standard error 0.1156 of Eq. 32.3.4: 5 per cent wider, which is the uncertainty about σ_a^2 that the plug-in throws away, exactly as [Proposition 32.6.1\(c\)](#) says. The same happens for the fixed effect: the posterior standard deviation of μ is 0.0709, against the plug-in $\sqrt{\hat{\lambda}_1/n} = 0.0672$.

Panel (b) of [Figure 32.6.1](#) repeats the analysis on the first 5 laboratories. The posterior medians are now 0.0435 and 0.0693, a difference of sixty per cent, against a REML estimate of 0.0493. With five groups the prior is doing a large part of the work, and saying so is not optional.

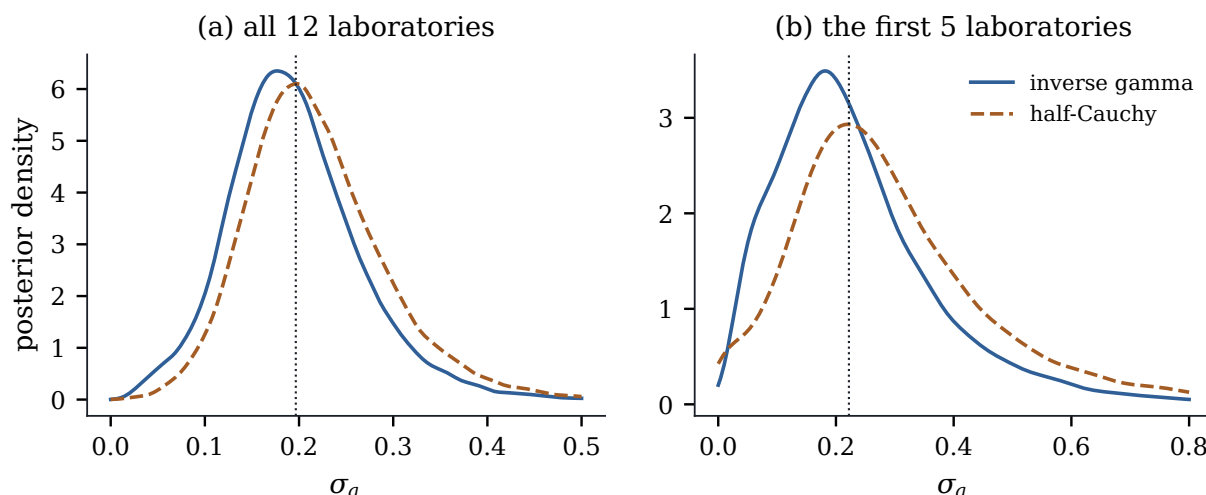


Figure 32.6.1: Posterior densities of the between-laboratory standard deviation σ_a under two priors, from a Gibbs sampler with 40 000 draws. The dotted line marks the REML estimate. (a) All twelve laboratories. (b) The first five, where the prior matters a great deal.

32.6.4 What the Bayesian version buys, and what it costs

Three gains. The uncertainty about $\boldsymbol{\theta}$ is carried through automatically, so intervals for fixed effects and prediction intervals for random effects are honestly wider, with no need for the corrections of [Section 32.5](#). The boundary problem disappears: the posterior of σ_a^2 is a proper distribution on $[0, \infty)$, with no point mass and no chi-bar-squared mixture to calibrate. And the same machinery handles models with no closed form, which is what [Chapter 33](#) needs for richer covariance structures and [Chapter 40](#) needs once the response is not normal.

Two costs. A prior for the variance components must be chosen, and with few groups it matters, as panel (b) shows; the only defensible practice is to state it and show the sensitivity. And the answers are not exactly the frequentist ones: the two frameworks agree on the point estimates, but they part company on the spread, and the frequentist coverage of a credible interval for a variance component is not guaranteed ([Proposition 12.6.4](#)).

32.6.5 Exercises

32.6.5.1 A. Check your understanding

Exercise A1.

Explain in two sentences why a random effect and a prior are formally the same object, and name one thing that distinguishes them in practice.

32.6.5.2 B. Practice

Exercise B1.

Show that in the one-way model the marginal posterior density of σ_a^2 under the prior $\pi(\sigma_a^2) \propto \sigma_a^{-2}$ is not integrable, by showing that $\exp\{\ell_R(\sigma_a^2)\}$ tends to a strictly positive limit as $\sigma_a^2 \downarrow 0$ with σ^2 profiled out. Contrast with the behaviour of the prior $\pi(\sigma_a) \propto 1$.

Solution. By Eq. 32.4.8, $-2\ell_R$ is continuous in (λ_1, λ_2) on $\lambda_1 \geq \lambda_2 > 0$, and at $\sigma_a^2 = 0$ it equals $(n-1)\log \lambda_2 + (\text{SSB} + \text{SSE})/\lambda_2$, finite and minimized at a positive λ_2 . So $\exp\{\ell_R\} \rightarrow c > 0$ as $\sigma_a^2 \downarrow 0$, and $\int_0^\delta c \sigma_a^{-2} d\sigma_a^2 = \infty$. With $\pi(\sigma_a) \propto 1$ the Jacobian gives $\pi(\sigma_a^2) \propto \sigma_a^{-1}$, and $\int_0^\delta \sigma_a^{-1} d(\sigma_a^2) = 2 \int_0^{\sqrt{\delta}} d\sigma_a < \infty$: the posterior is proper near the origin.

Exercise B2.

In Proposition 32.6.1(c), suppose $\boldsymbol{\theta}$ takes only two values with posterior probabilities 1/2 each, giving conditional posterior means $\hat{\mathbf{u}}_1, \hat{\mathbf{u}}_2$ and common conditional variance v for a coordinate. Compute the correct marginal posterior variance, and say how large the disagreement between $\hat{u}_1$ and $\hat{u}_2$ must be before the plug-in understates the standard deviation by ten per cent.

32.6.5.3 C. Going deeper

Exercise C1.

Formula Eq. 32.6.2 shows REML is the posterior mode of $\boldsymbol{\theta}$ under a flat prior on $\boldsymbol{\theta}$. A mode is not invariant to reparameterization: the mode of the posterior of σ_a^2 is not the square of the mode of the posterior of σ_a . Reconcile this with Theorem 32.4.2(c), which says the REML estimate is well defined. What is invariant, and what is not?

Exercise C2.

Take the balanced one-way model with σ^2 known, $m = 1$ so that $\bar{y}_k = y_k$, and μ known to be zero. Show that plugging the unbiased moment estimate of σ_a^2 into Eq. 32.3.3 gives a shrinkage factor of the form $1 - c\sigma^2/\sum_k y_k^2$, and compare with the James–Stein estimator of Theorem 27.5.2. Which constant c does the plug-in give, and which does James–Stein use?

Solution. With $m = 1$ and $\mu = 0$ the data are $y_k \sim N(a_k, \sigma^2)$ with $a_k \sim N(0, \sigma_a^2)$, so marginally $y_k \sim N(0, \sigma^2 + \sigma_a^2)$. The moment estimate from $\sum_k y_k^2$ is $\hat{\sigma}_a^2 = \sum_k y_k^2/g - \sigma^2$, and

$$B = \frac{\hat{\sigma}_a^2}{\hat{\sigma}_a^2 + \sigma^2} = 1 - \frac{g\sigma^2}{\sum_k y_k^2},$$

so $c = g$ in the stated form. The James–Stein estimator of Theorem 27.5.2 uses $1 - (g-2)\sigma^2/\sum_k y_k^2$. The difference is the same degrees-of-freedom correction that separates ML from REML: $g-2$ is the constant that makes $(g-2)\sigma^2/\sum_k y_k^2$ unbiased for $\sigma^2/(\sigma^2 + \sigma_a^2)$, and it is what makes the estimator dominate the unshrunk one for every $\boldsymbol{\alpha}$, not merely on average.